

bit encryption technology. Data storage encryption of sensitive fields is done. Any Proposal document that is uploaded to the server is subjected to symmetric encryption using a system generated symmetric key. Further this key is subjected to asymmetric encryption using buyers/Proposal opener's public keys. Overall, the uploaded tender documents become readable only after the tender opening by the authorized Proposal openers.

- h) The uploaded tender documents become readable only after the tender opening by the authorized Proposal openers.
- i) Upon the successful and timely submission of Proposals (i.e. after Clicking "Freeze Proposal Submission" in the portal), the portal will give a successful Proposal submission message & a Proposal summary will be displayed with the Proposal no. and the date & time of submission of the Proposal with all other relevant details.
- j) The Proposal summary has to be printed and kept as an acknowledgement of the submission of the Proposal. This acknowledgement may be used as an entry pass for any Proposal opening meetings.

ASSISTANCE TO CONSULTANTS

- a) Any queries relating to the tender document and the terms and conditions contained therein should be addressed to the Tender Inviting Authority for a tender or the relevant contact person indicated in the tender.
- b) Any queries relating to the process of online Proposal submission or queries relating to CPP Portal in general may be directed to the 24x7 CPP Portal Helpdesk.

Consultants may avail the free training on the use of e-procurement system as per the schedule published at the following link: <https://eprocure.gov.in/cppp/trainingdisp>. In case of any further queries, please contact the helpdesk available on the portal during office hours i.e. between 10 AM till 6 PM on weekdays.

7. Annexure 2- Functional Required Specifications

8. Introduction

The IMO Strategic Engagement Portal is a secure, web-based collaboration platform which is designed to streamline and digitize India's internal coordination process for international maritime engagements, particularly those under the International Maritime Organization (IMO), the International Labour Organization (ILO), and the International Maritime Satellite Organization (IMSO).

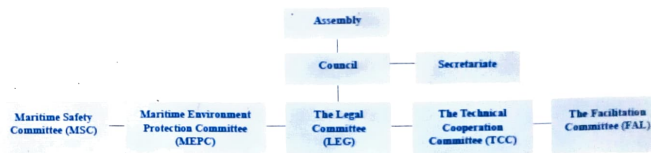
Built to support India's growing influence in global maritime governance, the portal enables seamless collaboration between technical experts, administrative officials, policy makers, and delegation leaders. It provides a structured, role-based environment for reviewing agenda documents, assigning tasks, compiling feedback, and preparing formal interventions.

The platform ensures that India's national maritime positions are:

- a) Well-informed through collective expert inputs
 - b) Timely in submission to international bodies
 - c) Digitally traceable with audit-ready workflows
 - d) Collaboratively developed with real-time task management and document control
- This platform will position DGS and India's maritime stakeholders for greater preparedness, participation, and policy shaping at international forums. It eliminates dependency on fragmented offline coordination and enables institutional memory by preserving all feedback, versions, and approvals in a structured digital archive.

Whether preparing for an IMO committee meeting, drafting amendments to the Maritime Labour Convention (MLC), or participating in a bilateral maritime dialogue, the IMO Shadow Committee Portal serves as a unified portal for India's global maritime diplomacy.

To carry out its wide-ranging mandate, the IMO is supported by a robust governance and technical structure, comprising an Assembly, a Council, and five principal technical committees, further assisted by specialized sub-committees.

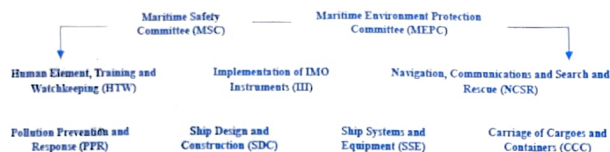


The technical work of the International Maritime Organization is carried out by five principal Committees. These include:

- a) The Maritime Safety Committee (MSC)
- b) The Marine Environment Protection Committee (MEPC)
- c) The Legal Committee
- d) The Technical Cooperation Committee, for capacity building
- e) The Facilitation Committee, to simplify the documentation and formalities required in international shipping. The Committees meet once or twice a year attended by Member States and NGOs

The MSC and MEPC are assisted in their work by several sub-committees which are open to all Member States. The committees are:

- Sub-Committee on Human Element, Training and Watchkeeping (HTW)
- Sub-Committee on Implementation of IMO Instruments (III)
- Sub-Committee on Navigation, Communications and Search and Rescue (NCSR)
- Sub-Committee on Pollution Prevention and Response (PPR)
- Sub-Committee on Ship Design and Construction (SDC)
- Sub-Committee on Ship Systems and Equipment (SSE)
- Sub-Committee on Carriage of Cargoes and Containers (CCC)



1. User Roles

The platform shall implement role-based access control (RBAC) with the following roles and privileges:

- Super User (DGS/DDG – International Cooperation):** Full configuration rights, including committee setup, user onboarding, workflow management, and oversight.
- Administrators:** Manage shadow committees, agenda, meetings, documents, and task assignments.
- Delegation Leaders:** Lead committee-level activities, consolidate feedback, approve final papers, and manage delegation outputs.
- Coordinators:** Manage sub-groups or correspondence groups, assign tasks, consolidate member feedback.
- Members:** Review agenda documents, provide comments/feedback, complete assigned tasks.
- Consultants/Support Staff:** Support drafting and documentation under delegated authority.

2. Functional Requirements

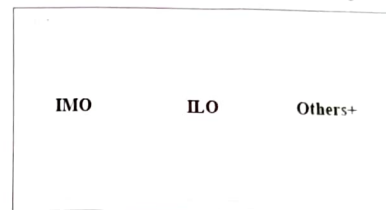
A. User Authentication and Security

- Secure login with username and password.
- Multi-factor authentication (MFA) for sensitive roles.
- Role-based permissions ensuring users only access relevant committees, documents, and tasks.
- Full audit trail of login attempts, approvals, edits, and submissions.
- Support for Single Sign-On (SSO) with government identity systems.

B. Shadow Committee and Meeting Management

- Repository of all IMO committees, sub-committees, Assembly, Council, with ability to expand for ILO, IMISO, and bilateral/regional collaborations (such as JWG).
- Meeting creation and scheduling with metadata (dates, location, agenda, participants)

- Dashboard view per meeting showing agenda, uploaded documents, tasks, deadlines, and participation status.
- Chronological listing of past and upcoming meetings.
- Calendar sidebar with hover-over summaries of upcoming meetings.



C. Agenda and Group Management

- Each meeting to contain agenda items with linked documents, tasks, and assigned members.
- Creation of groups and sub-groups linked to agenda items.
- Group roles: Convener, Coordinator, Members.
- Ability to create correspondence groups with dedicated task tracking.
- Automated role-based notifications and reminders for assigned responsibilities.
- Self-service for admins to add consultants/support staff.

D. Task Management

- Task creation by leaders or coordinators with linkage to agenda/document.
- Assignment to one or multiple members with due dates.
- Member view of pending, ongoing, and completed tasks via personal dashboard.
- Automatic reminders for approaching or overdue deadlines.
- Escalation workflow for unresolved tasks.
- Leaders' dashboard summarizing all pending/completed tasks. Task completion summary reports exportable in XML/Excel formats.

Sn	Agenda and Source	Name of the Document	Summary	Comments	Concerned person/group
1					
2					
3					

E. Document and Paper Preparation

- Upload, view, download, and archive official meeting documents.
- Creation of dedicated paper preparation folders for each meeting/agenda.
- Multi-level approval workflow: Member → Coordinator/group leader → Delegation Leader → CS/NA/CSS → DG/MoPSW.
- Version control with automatic timestamping and rollback capability.
- Document locking upon finalization by Delegation Leader.
- Export options in PDF, XML, and Excel formats.

F. Feedback and Review

- Members provide structured feedback via rich-text editor.
- Feedback consolidated by coordinators at group level.



- c) Delegation leader compiles committee-level final response
- d) Dashboards showing pending/overdue feedback at user, group, and committee level.
- e) Historical archive of feedback per meeting/agenda for institutional memory
- f) Automated tools for comparison of successive versions
- g) Structured templates for interventions, comments, and national positions.
- h) AI-assisted summarization of multiple feedback inputs (optional, future-ready).

G. Collaboration and Discussion Interface

- a) Real-time discussion boards linked to specific agenda items
- b) Ability for multiple users to simultaneously edit draft papers with attribution and tracked changes
- c) Commenting, tagging, and @mention functionality for collaboration
- d) Final approval workflow with locking of agreed text before submission.

H. Reporting and Monitoring

- a) Auto-generation of Summary Report, Minutes of Meetings (MoM) with attendee list, agenda discussion, and decisions taken after the IMO meeting.
- b) Preparation status reports (per committee, per meeting).
- c) Tracking of national stand/intervention across agenda items.
- d) Member performance metrics: meetings attended, agenda involvement, interventions made
- e) Visualization tools (charts, progress indicators) for leadership monitoring.
- f) Export options: PDF, Excel, XML.

I. International Engagements Module ("Others +")

- a) Repository for bilateral and regional working group activities.
- b) Upload of Terms of Reference, agreements, MoUs, minutes, and action items.
- c) Tracking of outcomes, resolutions, and progress against commitments.
- d) Summary reports for download and sharing with stakeholders.

3. Non-Functional Requirements

The IMO Strategic Platform shall comply with the following non-functional requirements to ensure reliability, security, scalability, and user experience for internal stakeholders of DGS, MoPSW, and MMDs.

A. Security

- a) Secure authentication using email/official ID-based login.
- b) Role-based access control ensuring data confidentiality and restricted permissions.
- c) Encrypted data transmission (HTTPS/TLS 1.3 or higher).
- d) Audit logs for all user actions including login, upload, edits, approvals, and feedback.
- e) Periodic security testing and compliance with MeitY Cloud Security Guidelines.

B. Usability

- a) Intuitive web-based interface accessible via standard browsers (Chrome, Edge, Firefox).
- b) Role-specific dashboards (Member, Coordinator, Delegation Leader, Administrator).
- c) Rich-text editor with track changes for collaborative document editing.
- d) Responsive design for desktops, laptops, and tablets.
- e) Context-sensitive help and tooltips for first-time users.

C. Performance and Scalability

- a) The system shall support a minimum of 150 concurrent users with scalable architecture to extend up to 300 concurrent users during peak activity periods.

- b) Response time for common transactions (login, document retrieval, feedback submission) shall not exceed 3 seconds under normal load.
- c) Upload/download of documents (up to 100 MB)
- d) Database design shall ensure optimized query performance for agenda, task, and document retrieval
- e) The system must be designed to scale horizontally with the increase in the user base (e.g. containerized or microservices-ready deployment)

D. Availability and Reliability

- a) Minimum 99.5% uptime excluding planned maintenance
- b) Planned maintenance windows must be notified at least 48 hours in advance
- c) Automated daily backups with retention of at least 90 days
- d) Disaster recovery provision with RPO ≤ 15 mins and RTO ≤ 2 hours
- e) Redundancy at application and database tiers to prevent single point of failure

E. Auditability and Compliance

- a) Complete audit trail of user activities, accessible only to administrators
- b) Compliance with GFR 2017, MeitY Cloud Guidelines, and ISO/IEC 27001 best practices.
- c) Logging of system errors and exceptions with proactive monitoring alerts

F. Interoperability

- a) Support for import/export of documents and reports in XML, Excel, and PDF formats
- b) Open APIs for potential future integration with DG Shipping E governance or other national maritime digital platforms.

4. Comprehensive Workflow of the Platform

A. User & Committee Setup

- a) Annual nomination of members (internal & external) by Administration
- b) Delegation Leader validates and approves external members
- c) Administrator creates committee structures (Assembly, Council, Committees, Sub Committees, Working Groups).
- d) User roles & access rights assigned (Leader, Convener, Coordinator, Member, Support Staff).

B. Agenda & Meeting Preparation

- a) Agendas of forthcoming meetings uploaded by Delegation Leader/Convener
- b) Papers linked to agenda items uploaded into Paper Preparation Folders
- c) Delegation of each agenda to responsible member(s)
- d) Leader of each agenda identified for ownership
- e) Automatic calendar updates & dashboard alerts for members

C. Task Distribution & Group Management

- a) Leaders distribute issues and tasks to agenda members
- b) Creation of Correspondence Groups or sub-groups for specialized issues
- c) Each task tagged with:
 - Agenda ID,
 - Responsible member(s),
 - Due date,
 - Linked documents.
- d) System-generated reminders for deadlines

D Document Preparation & Collaboration

- Members draft papers/interventions using collaborative editor
- Features Track changes, versioning, comments, real-time editing
- Drafts go through sequential workflow
Member → Group Leader → Delegation Leader → CS/NA/CSS → DG Shipping → MoPSW
- Final draft locked after approval

E Feedback & Consolidation

- Members provide feedback on agenda-linked documents
- Group Leaders consolidate feedback from their groups
- Delegation Leaders prepare final position papers (India's stand on each agenda)
- Feedback/stand documents exported in Excel/XML/PDF formats

F Pre-Meeting Approval

- Final stand including interventions and draft papers reviewed by Delegation Leaders
- Approval workflow Delegation Leaders → DG Shipping → Ministry (MoPSW)
- Approved positions locked for meeting use

G Meeting Phase (Dynamic Updates) *

- Real-time intervention edits made as discussions evolve
- Platform supports live co-editing and chat/discussion board
- Any changes tracked with timestamp and author

H Post-Meeting Documentation

- Final approved output on each issue uploaded by agenda leaders
- Leader of Delegation gives final approval
- Automatic generation of summary reports (issue-wise, intervention-wise, action points)

I To-Do List & Action Tracking

- System auto-creates a To-Do List of pending regulations, amendments, and follow-ups
- Assigned to Convenor and approved by Leader of Delegation
- Deadlines and progress tracked via dashboard

J Reporting & Monitoring

- Leaders and Admins generate reports anytime
 - Status of agenda deliberations
 - To-do item tracking
 - Committee performance dashboards
 - Member participation reports
- Reports downloadable in XML, Excel, PDF

K Archival & Knowledge Management

- Completed meeting cycles archived with:
 - Meeting ID, Agenda reference, Date, Contributors
- Searchable archive by committee, keyword, agenda ID
- Past interventions and position papers reusable as references

9. Annexure 3- Contract Agreement Form

This CONTRACT AGREEMENT (hereinafter called the "Contract") is made and entered into on this the [Insert Day] day of [Insert Month], 20*[Insert Year]** at [Insert City], India

BETWEEN

Directorate General of Shipping (client), Ministry of Ports, Shipping and Waterways, Government of India, having its office at 9th Floor, Beta Building, I-Think Techno Campus, Kanjurmarg (East), Mumbai - 400 042 (hereinafter referred to as the "client" or "client", which expression shall, unless repugnant to the context or meaning thereof include its successors and assigns) of the First Part

AND

[Name of Consultant], a company incorporated under the Companies Act, 1956/2013, having its registered office at [Address of Consultant] (hereinafter referred to as the "Consultant", which expression shall, unless repugnant to the context or meaning thereof include its successors and assigns) of the Second Part
(client and the Consultant are hereinafter referred to individually as "Party" and collectively as the "Parties")

WHEREAS:

- The client has issued a Request for Proposals (RFP) for the provision of certain consulting services as defined in this Contract (hereinafter called the "Services").
- The Consultant, having represented to the client that it has the required professional skills, expertise and technical resources, has submitted a Consultant's Proposal and has been selected to provide the Services on the terms and conditions set forth in this Contract
- The Consultant has agreed to provide the Services under the terms and conditions set forth herein.

NOW, THEREFORE, in consideration of the mutual covenants, promises, and agreements contained herein, the Parties hereto agree as follows:

1. Integral Documents

The following documents attached hereto shall be deemed to form and be read as an integral part of this Contract:

- The General Conditions of Contract (GCC);
- The Special Conditions of Contract (SCC);
- The Appendices:

* Appendix A: Terms of Reference (Scope of Work, Deliverables, and Timelines)

* Appendix B: Key Experts

- UI/UX wireframes and user journeys
- iii. Define role-based access control, workflows, audit trails, dashboards, and reporting structures

The System Integrator (SI) shall design, implement, and finalise a secure, scalable, cloud-native, and modular system architecture. The architecture shall be approved by the Directorate General of Shipping (DGS) prior to development and shall meet the following minimum requirements.

- A. The frontend shall provide a responsive, secure, and user-friendly interface for all user roles.
 - i. Web-based application accessible through standard modern browsers (Chrome, Edge, Firefox).
 - ii. Responsive design supporting desktop, tablet, and mobile access.
 - iii. Role-based UI rendering (Admin, Delegation Leader, Coordinator, Member, Viewer).
 - iv. Support for accessibility guidelines (WCAG 2.1 Level AA).
 - v. Secure session management and inactivity time-outs.
 - vi. Internationalization support (future-ready).
 - vii. Separation of UI and business logic (thin client architecture).
 - viii. Protection against common vulnerabilities (XSS, CSRF, clickjacking).
- B. The backend shall handle business logic, workflows, data processing, and security enforcement.
 - i. Service-oriented or microservices-based architecture.
 - ii. Stateless application design for scalability.
 - iii. Role-based access enforcement at API level.
 - iv. Workflow engine to support multi-level approvals and task routing.
 - v. Centralized exception handling and logging.
 - vi. High availability and horizontal scalability.
- C. The middleware shall enable secure and controlled integration between system components and external systems.
 - i. API-first architecture using RESTful APIs.
 - ii. Secure API Gateway with authentication, throttling, and logging.
 - iii. Support for future integration with IMO, ILO, IMSO, or other Government systems.
 - iv. Message-based integration (where required) for asynchronous processing.
 - v. Versioned APIs to ensure backward compatibility.
- D. The data layer shall ensure integrity, availability, confidentiality, and long-term retention of critical information.
 - i. Enterprise-grade relational database.
 - ii. Data normalization and indexing for performance.
 - iii. Support for structured and unstructured data (documents, feedback, logs).
 - iv. Role-based data access control.
 - v. Encryption at rest and in transit.
 - vi. Point-in-time recovery and automated backups.
 - vii. Archival and retention policies aligned with Government norms.
- E. Logging, Monitoring & Audit Architecture
 - i. Centralized logging across all components.
 - ii. Real-time monitoring of application and infrastructure health.
 - iii. Audit trails for document changes, approvals, and user actions.
 - iv. Dashboard for system administrators.
 - v. Log retention as per Government norms.
- F. Architecture Governance

- Request for Proposal for the Selection of System Integrator for the Development of an IMO Strategic Engagement Platform
- i. The SI shall submit detailed architecture diagrams and documentation.
- ii. Any deviation from the approved architecture shall require prior written approval from DGS.
- iii. Architecture shall be reviewed at design, pre-Go-Live, and post-Go-Live stages.

3.3 Design & Development of the Application

The System Integrator (SI) shall be responsible for the end-to-end design, development, configuration, customization, and implementation of the complete IMO Strategic Platform, in accordance with the approved Functional Requirements Specification (FRS) Software/System Requirements Specification (SRS), and system design documents.

The SI shall ensure that the application is developed using a modular, scalable, and cloud-native architecture, enabling ease of maintenance, future enhancements, and seamless integration with other Government systems, international portals, and digital platforms as required by DGS.

The SI shall implement all business workflows, approval hierarchies, collaboration mechanisms, and reporting frameworks, ensuring accuracy, traceability, and compliance with Government usability, accessibility, and security standards.

The platform shall include, but not be limited to, the following integrated functional modules and capabilities:

- A. Committee, Sub-Committee, Working Group & Meeting Management
 - i. Configuration and management of IMO Assembly, Council, Committees, Sub-Committees, Working Groups, and other international forums (including ILO IMSO, and bilateral/multilateral engagements).
 - ii. Creation, scheduling, and management of meetings with participant mapping and role assignment.
 - iii. Agenda creation and assignment at meeting, committee, and group levels.
- B. Agenda, Document & Version Management
 - i. Secure upload, categorization, and storage of agenda papers, working documents, submissions, and reference materials.
 - ii. Document tagging based on committee, agenda item, source, and meeting.
 - iii. Version control with complete audit trail and time stamping of all uploads and revisions.
 - iv. Secure in-system document viewing and controlled downloads.
- C. Collaboration, Feedback & Deliberation Management
 - i. Structured user-wise and group-wise feedback submission for agenda items and documents.
 - ii. Commenting, interventions, and deliberation notes with role-based visibility.
 - iii. Consolidation of inputs at multiple levels, enabling informed decision-making.
 - iv. Defined review workflow:
Member → Group Leader → Delegation Leader → IC Division.
- D. Task Allocation & Workflow Automation
 - i. Creation and assignment of tasks linked to agenda items, documents, and meetings.